

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 2.1: Plant Nutrition — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 2.1: Plant Nutrition
Driving Question	DRIVING QUESTION: Why does our sukuma wiki wilt in the midday sun even when the soil is wet — and what does a plant actually need to stay green, grow, and make its own food?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Why Does Our Sukuma Wiki Wilt? Anchoring the Phenomenon and Opening Our Questions	We observed photographs showing sukuma wiki wilting at midday despite moist soil, recovering overnight, and turning pale when moved into shade under a mango tree in a Nairobi kitchen garden	We learned that green plants are autotrophs — they manufacture their own food from simple inorganic raw materials — while organisms like mushrooms are heterotrophs that depend on ready-made organic food from other sources	We cannot yet fully explain why the sukuma wiki wilts at midday or why it turns pale in shade, but we have recorded our starting predictions and questions on the Driving Question Board and will return to them with evidence in every lesson that follows
Lesson 2 Leaf Structure: The Architecture of a Photosynthetic Machine	Students examined the internal cross-section of a dicot leaf (mango or bean) using a hand lens and microscope, identifying the upper epidermis, cuticle, palisade mesophyll, spongy mesophyll, air spaces, lower epidermis, stomata with guard cells, and vascular bundles (xylem and phloem).	Each leaf layer is structurally adapted to a specific function: the transparent cuticle/epidermis admits light while reducing water loss; the densely packed palisade mesophyll maximises chloroplast exposure to light; the spongy mesophyll provides air spaces for CO ₂ and O ₂ diffusion; stomata regulate gas exchange and water vapour loss; xylem transports water and minerals upward while phloem transports sugars downward.	The sukuma wiki wilts at midday even in moist soil because strong light triggers stomata to open wide for gas exchange (CO ₂ in, O ₂ out), simultaneously allowing rapid water vapour loss (transpiration) through the lower epidermis. When the rate of water vapour loss through open stomata exceeds the rate at which xylem can deliver water from the roots, leaf cells lose turgor pressure and the plant wilts. At night, stomata close, transpiration stops, xylem catches up, turgor is restored, and the plant recovers — explaining the daily wilting-and-recovery cycle.
Lesson 3 Raw Materials for Photosynthesis: What Does a Plant Actually	We read evidence showing that carbon dioxide enters the leaf through stomata, water travels from the soil up the xylem to mesophyll cells, light energy from the sun	Photosynthesis requires four raw materials: carbon dioxide, water, light energy, and chlorophyll. The word equation is: carbon dioxide plus water in the presence of light	The sukuma wiki wilts at midday because peak solar irradiance causes stomata to open wide for CO ₂ uptake, which simultaneously increases transpiration so

Need?	peaks in Nairobi between 11:00 and 14:00 coinciding with wilting, and chlorophyll in the palisade mesophyll absorbs that light.	energy and chlorophyll yields glucose plus oxygen. The balanced chemical equation is: 6CO_2 plus $6\text{H}_2\text{O}$ yields $\text{C}_6\text{H}_{12}\text{O}_6$ plus 6O_2 . Chlorophyll is not consumed in the reaction but is essential as a light-absorbing pigment.	water leaves the leaf faster than the xylem can replace it — even when soil is moist. The pallor and slow growth in shade are explained by reduced light energy reaching chlorophyll, which reduces the rate of photosynthesis and therefore reduces glucose production, slowing growth and reducing the stimulus for chlorophyll synthesis.
Lesson 4 Starch Test Practical: Finding Evidence That Chlorophyll and Light Are Both Needed for Photosynthesis	When iodine solution was applied to a decolourised variegated or partially covered leaf: the white region (no chlorophyll) remained brown-orange; the covered green region (chlorophyll present but no light) remained brown-orange; the green uncovered region (chlorophyll present and light available) turned blue-black, indicating starch was present only in that zone.	Both chlorophyll AND light must be present simultaneously for photosynthesis to occur and for starch to be produced in a leaf. Removing either one — even if the other is present — is sufficient to prevent photosynthesis. This is direct experimental evidence, not just a stated fact.	The shaded sukuma wiki in the phenomenon turns pale and grows slowly because, even though it retains chlorophyll, reduced light intensity under the mango tree limits photosynthesis and therefore reduces the glucose and starch available for growth — consistent with the starch test result showing that the covered green zone (chlorophyll present, light absent) produces no starch.
Lesson 5 Light-Dependent Reactions: Where the Sun's Energy Enters the Plant	Learners reviewed Elodea oxygen-bubble count data from Lesson 4 and examined a half-labelled thylakoid membrane diagram with a set of 6 reaction-event cards describing the steps of the light-dependent reactions.	Learners traced the sequence of light-dependent reactions: light absorption by chlorophyll in PSII and PSI → excitation and transfer of electrons → photolysis of water ($2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$) as the electron source → chemiosmotic synthesis of ATP (photophosphorylation) via ATP synthase → reduction of NADP^+ to NADPH at PSI → release of ATP and NADPH to the stroma for use in the Calvin cycle. They connected O_2 release to the Elodea bubble observations and connected reduced ATP/NADPH production under shade to the pale, slow-growing sukuma wiki.	More light hitting the sukuma wiki leaf means more chlorophyll molecules are excited in Photosystem II, driving faster photolysis of water, faster electron flow along the ETC, and greater production of ATP and NADPH — which is why the Elodea produces more bubbles at higher light intensities. When the sukuma wiki is moved into the shade of the mango tree, the reduced light energy means less chlorophyll excitation, slower photolysis, and therefore less ATP and NADPH produced — starving the Calvin cycle of the energy and reducing power it needs to fix CO_2 into sugars, which explains the slow growth and yellowing (pallor) of the shaded plant.
Lesson 6 Light-Independent Reactions: How the Calvin Cycle Builds Glucose from CO_2	Learners traced the colour-coded Calvin cycle diagram (carbon in red, ATP use in blue, NADPH use in green) and physically sequenced 10 laminated cards — including ' CO_2 fixed onto RuBP by RuBisCO enzyme', 'ATP and NADPH from light reactions arrive in stroma', and 'Less light → less ATP/NADPH → cycle slows → less glucose → slower growth' — to reconstruct	Learners learned that the light-independent reactions (Calvin cycle) occur in the stroma of the chloroplast, where CO_2 is fixed onto RuBP by RuBisCO to form 3-PGA, which is then reduced to G3P using ATP and NADPH imported from the light-dependent reactions; G3P is used to synthesise glucose and to regenerate RuBP. Less light reduces ATP and NADPH production,	Using a collaboratively constructed two-stage flow diagram, learners explained the complete causal chain linking reduced sunlight under the mango tree to slow growth in sukuma wiki: less light → fewer electrons excited in thylakoid → less ATP and NADPH produced → Calvin cycle cannot efficiently reduce 3-PGA to G3P → less glucose synthesised → fewer carbon

	the full light-independent reaction pathway in the context of Amina's shaded sukuma wiki.	slowing the Calvin cycle and producing less glucose, which explains why shaded sukuma wiki grows more slowly despite not wilting.	skeletons and less energy for cell division and growth → sukuma wiki grows slowly and eventually loses chlorophyll pigmentation.
Lesson 7 Factors Affecting the Rate of Photosynthesis — Planning & Carrying Out an Investigation	Learners counted oxygen bubbles produced by an aquatic plant (or read a simulated dataset) at five different light intensities (lamp distances: 5 cm, 10 cm, 20 cm, 30 cm, 40 cm). They recorded bubble counts per minute in a results table and plotted a line graph of rate of photosynthesis (bubbles min ⁻¹) vs light intensity.	The rate of photosynthesis increases as light intensity increases, but only up to a maximum — above that point the curve flattens, indicating that another factor (CO ₂ concentration or temperature) has become the limiting factor. A LIMITING FACTOR is the factor present at the lowest level that controls the rate of a process. The pale, slow-growing sukuma wiki under the mango tree is in the low-light-intensity region of the graph where photosynthesis rate is low.	Why the sukuma wiki under the mango tree grows slowly and turns pale: the shade reduces light intensity, placing the plant in the region of the rate–intensity graph where photosynthesis rate is low. Insufficient glucose is produced for growth, and insufficient energy is available for chlorophyll synthesis, causing pallor. However, the wilting puzzle remains unexplained by this graph — water movement out of the leaf is a missing piece that will be addressed in the next lesson.
Lesson 8 Factors Affecting Photosynthesis Rate — Data Analysis & Explain	Learners plotted and annotated line graphs showing how CO ₂ concentration and temperature each affect the rate of photosynthesis, and compared these with their Lesson 7 light-intensity graphs.	The rate of photosynthesis is controlled by limiting factors — light intensity, CO ₂ concentration, and temperature — and the factor in shortest supply at any moment determines the overall rate. Temperature graphs show a bell-curve shape because enzymes denature above ~40 °C.	Learners used the limiting factor concept to explain why the sukuma wiki under the mango tree grew pale and slowly (low light was the limiting factor), why greenhouse tomato farmers in Naivasha add CO ₂ to boost yields (CO ₂ was the limiting factor in well-lit, warm greenhouses), and why adding CO ₂ to cool, misty Kericho tea farms would not significantly increase the photosynthesis rate (temperature, not CO ₂ , is limiting there).
Lesson 9 Mineral Nutrition: What Does the Soil Give the Plant?	Learners analysed KALRO deficiency photographs and secondary data tables showing mineral deficiency symptoms in Kenyan crops including maize from Nakuru, tea from Kericho, beans from Kisii, and sukuma wiki from Nairobi.	Plants require macronutrients (N, P, K, Ca, Mg, S) and micronutrients (Fe, Zn, etc.) absorbed from soil water through root hair cells by active transport; each mineral has a specific role in plant structure and metabolism, and deficiency produces characteristic visible symptoms such as chlorosis, necrosis, stunted growth, and purpling.	The pale yellowing of the sukuma wiki under the mango tree is partly explained by reduced photosynthesis due to low light, but is also consistent with nitrogen and magnesium deficiency causing chlorosis — insufficient mineral uptake means the plant cannot synthesise adequate chlorophyll, proteins, or new cell structures, so it grows slowly and its leaves turn pale even when soil water is available.
Lesson 10 Model Building, Synthesis and DQB Resolution: Putting the Sukuma Wiki Story Together	We revisited the original sukuma wiki wilting and pallor phenomenon and reviewed evidence gathered across nine lessons through annotated predictions, an Evidence Harvest gallery walk, and group annotated models integrating leaf structure, photosynthesis stages, limiting factors, and	The wilting of sukuma wiki at midday — even when soil is wet — occurs because high light intensity and temperature accelerate transpiration through open stomata faster than roots can absorb and transport water upward through xylem, causing loss of turgor pressure in leaf cells;	By integrating leaf anatomy, the two-stage mechanism of photosynthesis, the concept of limiting factors, and the roles of soil minerals, we can fully resolve the driving question: the sukuma wiki plant needs light as an energy source for the light-dependent reactions, water and CO ₂ as raw materials

	mineral nutrition.	this is a reversible process that reverses at night when transpiration slows and turgor is restored. The pallor and slow growth in shade occur because reduced light intensity limits the light-dependent reactions, reducing ATP and NADPH production, which in turn limits the Calvin cycle output of glucose; with less glucose available for growth and less chlorophyll synthesised (especially if nitrogen or magnesium are also limiting), the plant grows slowly and appears pale yellow-green.	for both stages, chlorophyll to absorb light, and minerals such as nitrogen and magnesium from the soil to build chlorophyll and enzymes — when any of these factors is limiting, the plant cannot make food efficiently, leading to the observable symptoms Amina saw in her Nairobi kitchen garden; understanding these mechanisms helps Kenyan smallholder farmers make evidence-based decisions about irrigation timing, shade management, and fertiliser application to improve sukuma wiki yield and contribute to household and national food security.
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ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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